

QINHEPING HU

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EMPLOYMENT

Bytedance Inc.

2025-present

Software Engineer

- Working on automatically generate unit tests using large language model. Knowledge in the theory of programming language, generative AI and prompt engineering is required to perform the job.

Amazon Inc.

2021-2025

Applied Scientist II @ Automated Reasoning Group (ARG)

- Developed loop-invariants support in the program verification tools CBMC (C bounded model checker) and Kani (verification for rust). Knowledge in the theory of programming language, formal method, and program verification is required to perform the job.
- Developed a program synthesizer to automatically generate loop invariants for Rust and C programs. Knowledge in program synthesis is required to perform the job.
- Worked on verification of the safety of the Rust standard library, which is the most popular library in Rust.
- Worked on introducing automated reasoning checks to the Large Language Model (LLM), which can help users generate a more accurate LLM response on highly regulated topics. Knowledge in generative AI and prompt engineering is required to perform the job.

EDUCATION

University of Wisconsin-Madison, Madison

2015-2021

Ph.D. in Computer Science, advised by Loris D'Antoni

Minor in Mathematics: Probability, Statistics, Stochastic Processes, and Category Theory

- My Ph.D. study is about extending a kind of popular synthesis problems (syntax-guided synthesis) with quantitative objectives and presenting an algorithm to solve such extended problems.

Tsinghua University, Beijing, China

2011-2015

B.E. in Computer Science, Special Pilot Class (Yao Class), advised by Prof. Giulio Chiribella

PUBLICATION

Programmable Program Synthesis **Invited keynote**

CAV 21

Jinwoo Kim, Qinheping Hu, Loris D'Antoni, Thomas Reps

Synthesis with Asymptotic Resource Bounds

CAV 21

Qinheping Hu, John Cyphert, Loris D'Antoni, Thomas Reps

Semantics-guided Synthesis

POPL 21

Jinwoo Kim, Qinheping Hu, Loris D'Antoni, Thomas Reps

Exact and Approximate Methods for Proving Unrealizability of Syntax-Guided Synthesis Problems

PLDI 20

Qinheping Hu, John Cyphert, Loris D'Antoni, Thomas Reps

Solving Program Sketches with Large Integer Values

ESOP 20

Rong Pan, Qinheping Hu, Rishabh Singh, Loris D'Antoni

Selected for special issue of TOPLAS

Nominated for EAPLS Award for the best ETAPS paper on PL and systems

Automatic Repair of Regular Expressions OOPSLA 19

Rong Pan, Qinheping Hu, Gaowei Xu, Loris D'Antoni

Direct Manipulation for Imperative Programs SAS 19

Qinheping Hu, Roopsha Samanta, Rishabh Singh, Loris D'Antoni

Proving Unrealizability for Syntax-Guided Synthesis CAV 19

Qinheping Hu, Jason Breck, John Cyphert, Loris D'Antoni, Thomas Reps

Syntax-Guided Synthesis with Quantitative Syntactic Objectives CAV 18

Qinheping Hu, Loris D'Antoni

Automatic Program Inversion using Symbolic Transducers PLDI 17

Qinheping Hu, Loris D'Antoni

Units of rotational information New J. Phys. 17

Yuxiang Yang, Giulio Chiribella, Qinheping Hu

WORKSHOP

Guarantees in Program Synthesis SYNT 19

Qinheping Hu, Jason Breck, John Cyphert, Loris D'Antoni, Thomas Reps

RESEARCH EXPERIENCE

Research Intern, Microsoft Research Summer 2017

- Designed and developed a framework to utilize NLP in fault localization and program repair. We took a subset of Microsoft Driver as our target, mutated them to produce more correct/buggy programs. And we used our model to decide whether a given program is buggy, where are the bugs, and what are possible fixes for the bugs.
- Mentor: Rishabh Singh.

Bachelor Thesis 2014-2015

- *Quantum Fission of Maximally Entangled Spins*
- This work is part of the paper Units of rotational information.

Semester Thesis Spring 2014

- *Survey on Partial Satisfaction* **Download**
- We survey the results of lower bounds of satisfied fraction in a k -satisfiable CNF formula.
- Advisors: Timon Hertli, Emo Welzl in ETH.

HONOR AND AWARD

Baidu Academic Scholarship 2011 - 2013

Tsinghua Scholarship for Special Pilot Class 2011 - 2015

Chinese High School Mathematics League · first prize 2010, 2011

SERVICE

Program Committee	VMCAI 2024, SYNT 2021
External Reviewer	FMCAD 2020
Artifact Evaluation Committee	PLDI 19

SKILLS

Language (Computer)	: C/C++ • Python • Rust • Shell
Language (Human)	: Chinese (native) • English (fluent)
Formal Method	: Program Verification • Program Synthesis • SMT Solver
Scientific Computing	: Numpy
Machine Learning	: Pytorch • statsmodels • pandas